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Python Inner Classes

Modularizing Complex Object Architectures

1. Introduction to Inner Classes

An **Inner Class** (also known as a Nested Class) is a class defined entirely within the body of another class.

They are used to group classes that are logically related but only relevant within the context of the container (Outer) class. This improves **Encapsulation** and readability.

2. Basic Syntax

To define an inner class, simply indent the new class definition inside the parent class.

```
class Outer:
    def __init__(self, name):
        self.name = name

    class Inner:
        def display(self):
            print("Action from Inner!")
```

3. External Instantiation

To create an object of an Inner class from outside, you must first have an instance of the Outer class.

```
outer = Outer()
# Accessing inner class through outer instance
inner = outer.Inner()
inner.display()
```

Alternatively: `inner = Outer().Inner()`

4. Accessing Outer from Inner

Inner classes do **not** automatically see the parent's data. You must explicitly pass the outer instance as a parameter.

```
class Outer:
    def __init__(self, val):
        self.val = val
        self.inner = self.Inner(self)

    class Inner:
        def __init__(self, outer_ref):
            self.outer = outer_ref
```

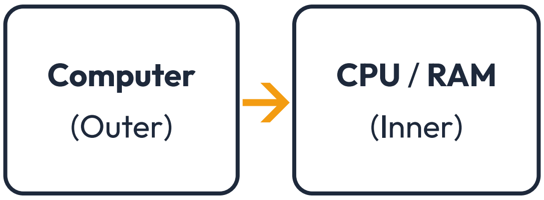
5. Practical Example: Car & Engine

A classic use case is a `Car` class containing an `Engine` inner class. Since an engine doesn't usually exist independently of a car, nesting makes sense.

The `Car` can manage the state of the `Engine`, and the `Engine` can store its specific performance metrics.

6. Multiple Inner Classes

A single outer class can manage multiple nested sub-systems, such as a `Computer` class housing `CPU` and `RAM` classes.



7. Organizing Large Classes

When a class becomes too large with too many variables, Inner classes help split the logic into manageable "sub-components" while keeping them under one namespace.

> This avoids "Flat" structures where 50 variables are all in one `__init__` method.

8. Inner Classes and Inheritance

Inner classes can inherit from other classes just like normal classes. They can also be inherited by other classes, providing a very flexible architectural depth.

> Note: Inheriting from an inner class requires referencing it via the outer class: `class Sub(Outer.Inner):`

9. Knowledge Check

Exercise

Question: What is an inner class in Python?

☐ A class that inherits from another class

☑ **A class defined inside another class**

☐ A class with private properties

10. Core Summary

- ✓ Defined within an Outer class.
- ✓ Used for logical grouping & organization.
- ✓ Instantiated through the Outer class.
- ✓ Does not see Outer data by default.
- ✓ Ideal for "Component-based" objects.

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Advanced Structural OOP Series